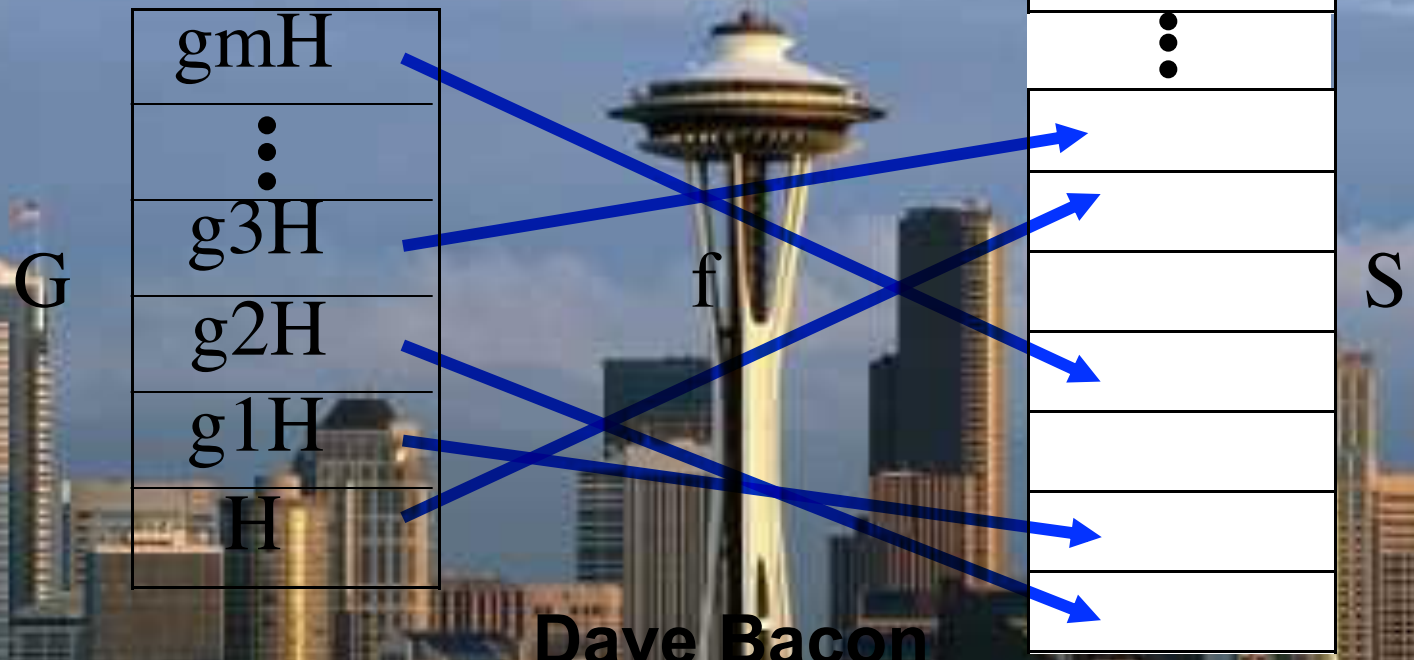
New Algorithms for the Nonabelian Hidden Subgroup Problem



Dave Bacon

Department of Computer Science & Engineering

University of Washington

in collaboration with Andrew Childs (Caltech) and Wim van Dam (UCSB)

Looking For New Algorithms

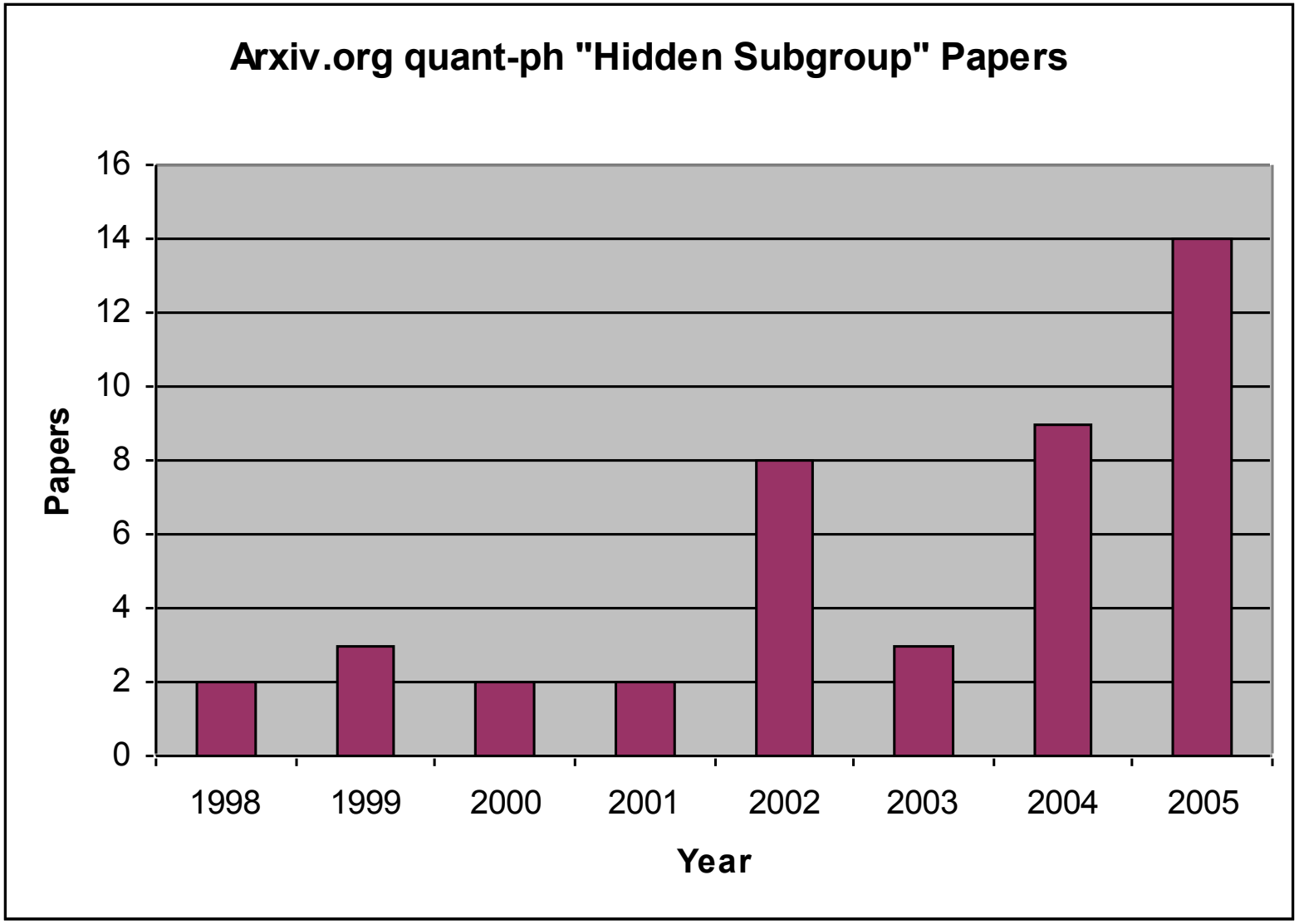
1996:

Quantum Computation and NP-Completeness

Dave Bacon
Summer Undergraduate Research Fellowship Paper
Advisors: Nicholas Cerf and Chris Adami
California Institute of Technology



Hidden Subgroup Problem



quant-ph/0501044 “Optimal measurements for the dihedral hidden subgroup problem,” Bacon, Childs, van Dam

quant-ph/0501056 “The Symmetric Group Defies Strong Fourier Sampling: Part I”, Moore, Russell, Schulman

quant-ph/0501177 “The Symmetric Group Defies Strong Fourier Sampling: Part II,” Moore, Russell

quant-ph/0501177 “For Distinguishing Conjugate Hidden Subgroups, the Pretty Good Measurement is as Good as it Gets”, Moore, Russell

quant-ph/0503095 “The Power of Strong Fourier Sampling: Quantum Algorithms for Affine Groups and Hidden Shifts,” Moore, Rockmore, Russell, Schulman

quant-ph/0503114 “On the Power of Random Bases in Fourier Sampling: Hidden Subgroup Problem in the Heisenberg Group” Radhakrishnan, Roetteler, Sen

quant-ph/0504067 “Explicit Multiregister Measurements for Hidden Subgroup Problems; or, Fourier Sampling Strikes Back,” Moore, Russell

quant-ph/0504083 “From optimal measurement to efficient quantum algorithms for the hidden subgroup problem over semidirect product groups,” Bacon, Childs, van Dam

quant-ph/0507190 “Quantum algorithm for a generalized hidden shift problem,” Childs, van Dam

quant-ph/0510185 “On the quantum hardness of solving isomorphism problems as nonabelian hidden shift problems,” Childs, Wocjan

quant-ph/0511054 “Strong Fourier Sampling Fails over Gn,” Alagic, Moore, Russell

quant-ph/0511148 “Limitations of Quantum Coset States for Graph Isomorphism,” Hallgren, Roetteler, Sen

quant-ph/0511149 “Tight Results on Multiregister Fourier Sampling: Quantum Measurements for Graph Isomorphism Require Entanglement,” Moore, Russell

quant-ph/0501044 “Optimal measurements for the dihedral hidden subgroup problem,” **Bacon**, **Childs**, **van Dam**

quant-ph/0501056 “The Symmetric Group Defies Strong Fourier Sampling: Part I”, **Moore**, Russell, **Schulman**

quant-ph/0501177 “The Symmetric Group Defies Strong Fourier Sampling: Part II,” **Moore**, Russell

quant-ph/0501177 “For Distinguishing Conjugate Hidden Subgroups, the Pretty Good Measurement is as Good as it Gets”, **Moore**, Russell

quant-ph/0503095 “The Power of Strong Fourier Sampling: Quantum Algorithms for Affine Groups and Hidden Shifts,” **Moore**, Rockmore, Russell, **Schulman**

quant-ph/0503114 “On the Power of Random Bases in Fourier Sampling: Hidden Subgroup Problem in the Heisenberg Group” Radhakrishnan, Roetteler, Sen

quant-ph/0504067 “Explicit Multiregister Measurements for Hidden Subgroup Problems; or, Fourier Sampling Strikes Back,” **Moore**, Russell

quant-ph/0504083 “From optimal measurement to efficient quantum algorithms for the hidden subgroup problem over semidirect product groups,” **Bacon**, **Childs**, **van Dam**

quant-ph/0507190 “Quantum algorithm for a generalized hidden shift problem,” **Childs**, **van Dam**

quant-ph/0510185 “On the quantum hardness of solving isomorphism problems as nonabelian hidden shift problems,” **Childs**, **Wocjan**

quant-ph/0511054 “Strong Fourier Sampling Fails over Gn,” Alagic, **Moore**, Russell

quant-ph/0511148 “Limitations of Quantum Coset States for Graph Isomorphism,” Hallgren, Roetteler, Sen

quant-ph/0511149 “Tight Results on Multiregister Fourier Sampling: Quantum Measurements for Graph Isomorphism Require Entanglement,” **Moore**, Russell

The Problem

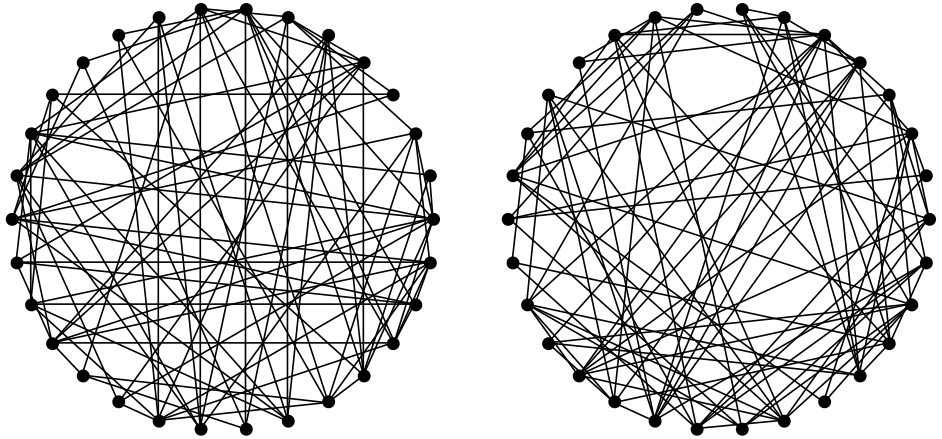
The Hidden Subgroup Problem (HSP)

Problem: Fix a group G (known) and a subgroup $H \subseteq G$ (unknown). Given query access to a function $f : G \to S$ that is constant and distinct on left cosets of H in G. We say that f hides H.

Goal: Find (a generating set for) H.

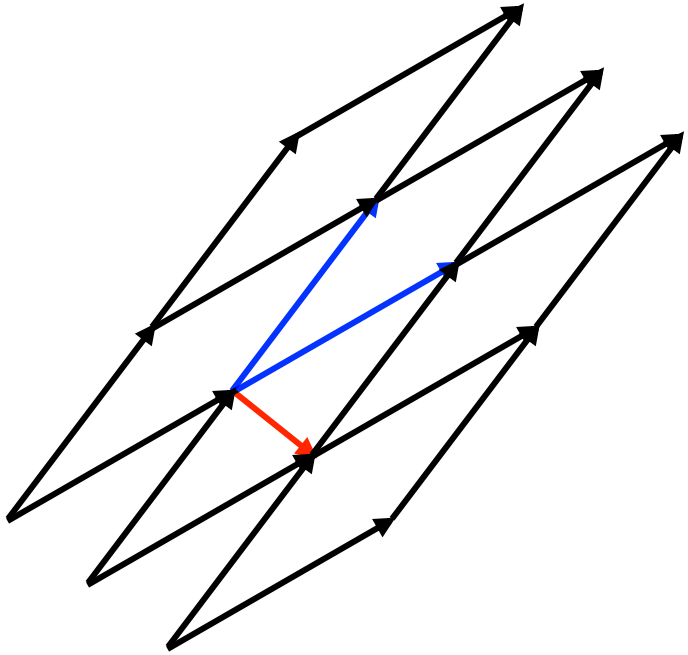
Efficient algorithm: run time $\text{poly}(\log |G|)$.

Why We Care



Graph Isomorphism

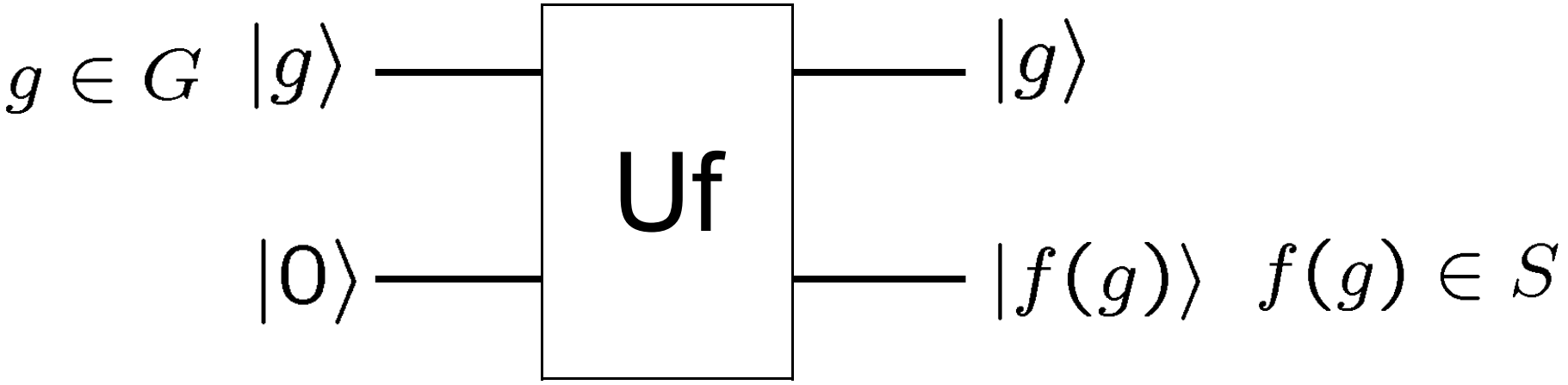
HSP for Symmetric Group



Unique k-Shortest Vector In a Lattice

HSP for Dihedral Group

Standard Query Model



$$\frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|G|}} |g\rangle|f(g)\rangle$$

constant and distinct on different left cosets of H:

$$\rho_H = \sum_{g \in \hat{G}} |gH\rangle\langle gH| \qquad |gH\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |gh\rangle$$

Hidden Subgroup State

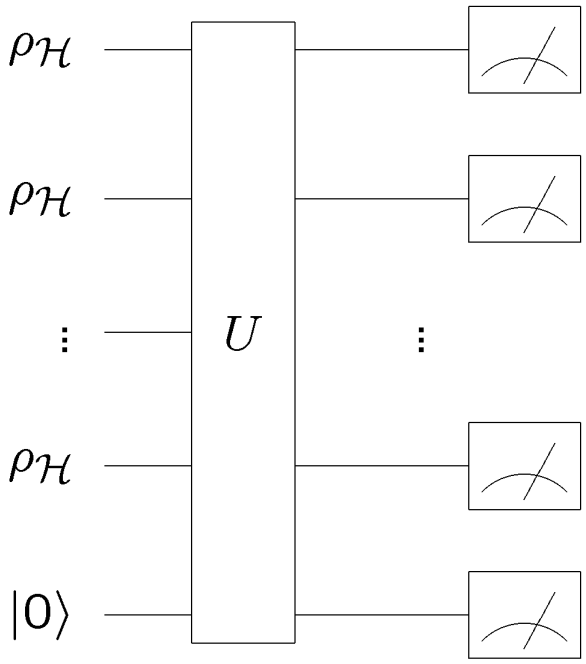
Copies

Single Copy

Given: $\rho_H = \sum_{g \in \hat{G}} |gH\rangle\langle gH|$

Multiple Copy

Given: $\rho_H^{\otimes k}$



Classic early result:

[Ettinger, Hoyer 98]

$O(\log |\mathcal{G}|)$ copies are sufficient
(but no efficient circuit known)

What Will It Look Like?

quant-ph/0501056 “The Symmetric Group Defies Strong Fourier Sampling: Part I”, Moore, Russell, Schulman

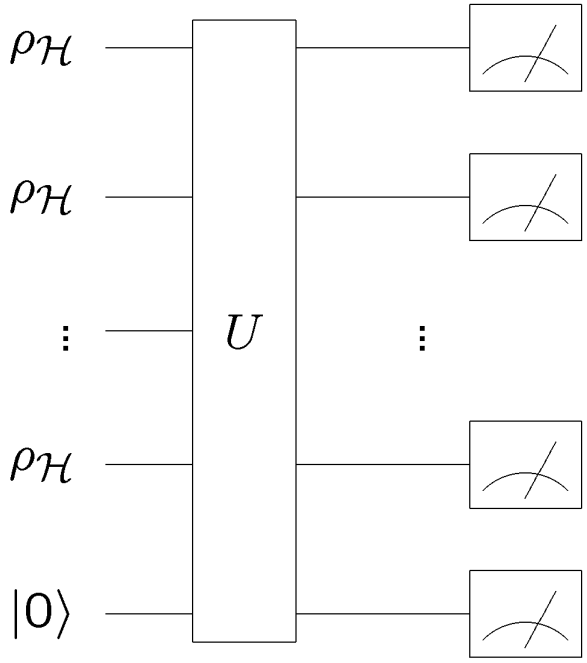
quant-ph/0501177 “The Symmetric Group Defies Strong Fourier Sampling: Part II,” Moore, Russell

quant-ph/0510185 “On the quantum hardness of solving isomorphism problems as nonabelian hidden shift problems,” Childs, Wocjan

quant-ph/0511148 “Limitations of Quantum Coset States for Graph Isomorphism,” Hallgren, Roetteler, Sen

quant-ph/0511149 “Tight Results on Multiregister Fourier Sampling: Quantum Measurements for Graph Isomorphism Require Entanglement,” Moore, Russell

Symmetric Group



Measurement on $\Omega(\log|\mathcal{G}|)$ copies are information theoretically necessary (and also sufficient.)

Nattering Nabobs of Negativism

"Our result can be viewed as a negative result because highly entangled measurements seem hard to implement in general."

- Hallgren, Roetteler, Sen
"Limitations of Quantum Coset States for Graph Isomorphism"

Is There Hope?

Bacon (UW), Childs (Caltech), and van Dam (UCSB)
quant-ph/0504083, FOCS 2005

HSP as State Distinction

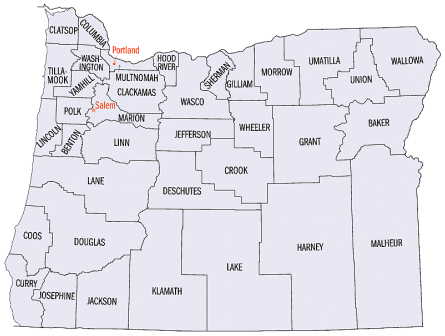
Single Copy

Given: $\rho_H = \sum_{g \in \hat{G}} |gH\rangle\langle gH|$ with probability *pH*

Determine: *H* by a generalized measurement

Maximize: probability of successfully identifying *H*





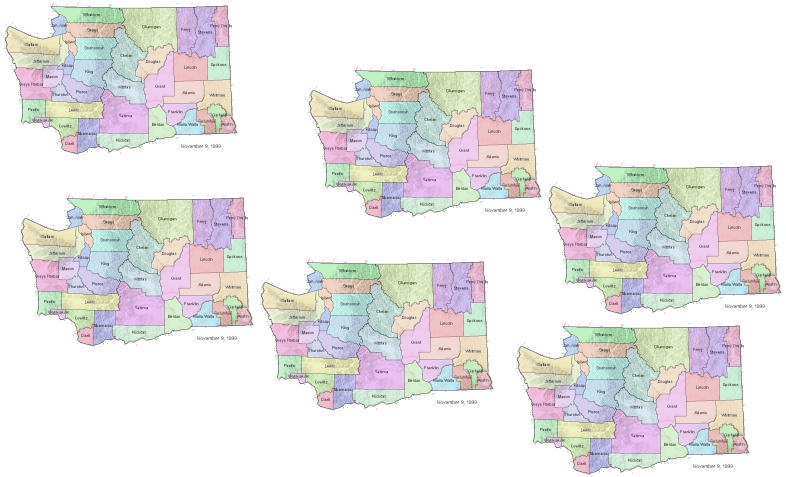
HSP as State Distinction

Multiple Copy

Given: $\rho_H^{\otimes k}$ with probability *pH*

Determine: *H* by a generalized measurement

Maximize: probability of successfully identifying *H*





Holevo's Condition

Given: ρ_i with probability *pi*

Measure: POVM with elements $\{G_j\}$

Outcome: *j* corresponds to guessing state ρ_j

Maximize: probability of successfully identifying ρ_j

Necessary and Sufficient Condition: [Holevo 73]

$$\sum_i p_i \rho_i G_i = \sum_i p_i G_i \rho_i \qquad \sum_i p_i \rho_i G_i \geq p_j \rho_j$$

- **Hard to find explicit forms for *Gj***

Pretty Good

Given: ρ_i with equal probabilities

Observation: the "Pretty Good Measurement" is often pretty good.

Pretty Good Measurement (PGM):

$$G_j = T^{-1/2} \rho_j T^{-1/2} \qquad T = \sum_i \rho_i$$

(plus $I - \sum_j G_j$ if needed.)

PGM is often optimal HSP Measurement

[Bacon, Childs, van Dam 05, Moore, Russell 05]

Heisenberg Group

Multiplication of upper triangular matrices over $\mathbb{Z}_p$

$$\begin{bmatrix} 1 & b & x \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \qquad x, y, b \in \mathbb{Z}_p$$



Heisenberg and Bohr

Margrethe Bohr to Heisenberg (from the play “Copenhagen”)

“Your talent is for ski-ing too fast for anyone to see where you are. For always being in more than one position at a time, like one of your particles.”

Heisenberg Group

midirect product group $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$



Elements of the group:

$$\left(\begin{pmatrix} x \\ y \end{pmatrix}, b \right) \quad x, y, b \in \mathbb{Z}_p$$

Multiplication rule:

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, b_1 \right) \left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_2 \right) =$$

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{b_1} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_1 + b_2 \right)$$

Heisenberg HSP

We can restrict to subgroups of the form

$$\mathcal{H}_{d_1,d_2} = \left\langle \left(\begin{pmatrix} d_1 \\ d_2 \end{pmatrix}, 1 \right) \right\rangle$$

$$\mathcal{H}_{d_1,d_2} = \left\{ \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0 \right), \left(\begin{pmatrix} d_1 \\ d_2 \end{pmatrix}, 1 \right), \left(\begin{pmatrix} 2d_1 + d_2 \\ 2d_2 \end{pmatrix}, 2 \right), \ldots \right\}$$

General version of theorem from [Ettinger, Hoyer 98]

Heisenberg HS State

Restricted subgroups:

$$\mathcal{H}_{d_1,d_2} = \left\{ \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0 \right), \left(\begin{pmatrix} d_1 \\ d_2 \end{pmatrix}, 1 \right), \left(\begin{pmatrix} 2d_1 + d_2 \\ 2d_2 \end{pmatrix}, 2 \right), \ldots \right\}$$

Random coset states:

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| \left(\left(\begin{pmatrix} l_1 \\ l_2 \end{pmatrix} + M^{(b)} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \right), b \right) \right\rangle$$

$$M^{(b)} = \sum_{i=0}^{b-1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^i = \begin{bmatrix} b & \frac{b(b-1)}{2} \\ 0 & b \end{bmatrix}$$

Coset labels:

$$l_1, l_2 \in_U \mathbb{Z}_p$$

Hidden Subgroup states:

$$\rho_{d_1,d_2} = \frac{1}{p^2} \sum_{l_1,l_2 \in \mathbb{Z}_p} |l_1, l_2, d_1, d_2\rangle\langle l_1, l_2, d_1, d_2|$$

Optimal Measumrent

The optimal measurement for k copies is again the pretty good measurement.

For one copy, the probability of successfully identifying the hidden subgroup is

$$Pr(succ) = O\left(\frac{1}{p}\right)$$

But for two copies, the probability of succesfully identifying the hidden subgroup is

$$Pr(succ) = O(1)$$

Can we efficiently implement this measurement?

HS States

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| \left(\left(\begin{pmatrix} l_1 \\ l_2 \end{pmatrix} + M^{(b)} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \right), b \right) \right\rangle$$

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \left| l_1 + bd_1 + \frac{b(1-b)}{2} d_2, l_2 + bd_2; b \right\rangle$$

Perform the QFT over $(\mathbb{Z}_p)^2$ to the first two registers:

$$|l_1, l_2, d_1, d_2\rangle = \frac{1}{\sqrt{p^3}} \sum_{b, x_1, y_1 \in \mathbb{Z}_p} \omega_p^{(l_1 + bd_1 + \frac{b(1-b)}{2} d_2) x_1 + (l_2 + bd_2) y_1} |x_1, y_1, b\rangle$$

Measure these two registers: $x_1, y_1 \in_U \mathbb{Z}_p$

$$|x_1, y_1, d_1, d_2\rangle = \frac{1}{\sqrt{p}} \sum_{b \in \mathbb{Z}_p} \omega_p^{(bd_1 + \frac{b(1-b)}{2} d_2) x_1 + bd_2 y_1} |b\rangle$$

HS States

Two copies: $x_1, y_1, x_2, y_2 \in_U \mathbb{Z}_p$

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{(b_1 x_1 + b_2 x_2) d_1 + \left(\frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2 \right) d_2} |b_1, b_2\rangle$$

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{v d_1 + w d_2} |b_1, b_2\rangle$$

Compute v and w:

$$\frac{1}{p} \sum_{b_1, b_2 \in \mathbb{Z}_p} \omega_p^{v d_1 + w d_2} |b_1, b_2\rangle |v, w\rangle$$

Uncompute

If we could erase the b1,b2 values:

$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle \rightarrow \frac{1}{p}|0,0\rangle\sum_{v,w}\omega_p^{vd_1+wd_2}|v,w\rangle$$

Then a QFT over $(\mathbb{Z}_p)^2$ would find d1 and d2.

Remember v and w are a function of b1 and b2.

Solving Equations

Compute v and w:

$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle$$

$$v = b_1x_1 + b_2x_2$$

$$w = \frac{b_1(1-b_1)}{2}x_1 + b_1y_1 + \frac{b_2(1-b_2)}{2}x_2 + b_2y_2$$

For fixed x1,x2,y1,y2 let $S_{v,w}$ be the set of b1,b2 solutions to these equations.

$$\frac{1}{p}\sum_{v,w\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}\left(\sum_{b_1,b_2\in S_{v,w}}|b_1,b_2\rangle\right)|v,w\rangle$$

We want to implement $\frac{1}{\sqrt{|S_{v,w}|}}\sum_{b_1,b_2\in S_{v,w}}|b_1,b_2\rangle|v,w\rangle \rightarrow |0,0\rangle|v,w\rangle$

Solving Equations

$$\frac{1}{\sqrt{|S_{v,w}|}} \sum_{b_1,b_2 \in S_{v,w}} |b_1, b_2\rangle |v, w\rangle \rightarrow |0, 0\rangle |v, w\rangle$$

Easier to think about the inverse (do it unitarily):

$$|0, 0\rangle |v, w\rangle \rightarrow \frac{1}{\sqrt{|S_{v,w}|}} \sum_{b_1,b_2 \in S_{v,w}} |b_1, b_2\rangle |v, w\rangle$$

i.e. given v and w, transform to superposition over solutions to the equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Solving Equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Assuming y1,y2,y1+y2 do not equal 0 (prob O(1/p)), then

$$\frac{p(p-1)}{2} \text{ solutions have } |S_{v,w}| = 0$$

$$p \text{ solutions have } |S_{v,w}| = 1$$

$$\frac{p(p-1)}{2} \text{ solutions have } |S_{v,w}| = 2$$

Solving Equations

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

Focus on when there are two solutions.

We can use standard finite arithmetic to find these solutions. Use to create labeled superposition of solutions:

$$\frac{1}{\sqrt{2}} \sum_{j=0,1} |j, (b_1)_j, (b_2)_j\rangle$$

Hadamard first qubit, measure. If 0, you have

$$\frac{1}{\sqrt{2}} \sum_{j=0,1} |(b_1)_j, (b_2)_j\rangle$$

Heisenberg HSP Solved

This allows us to perform:

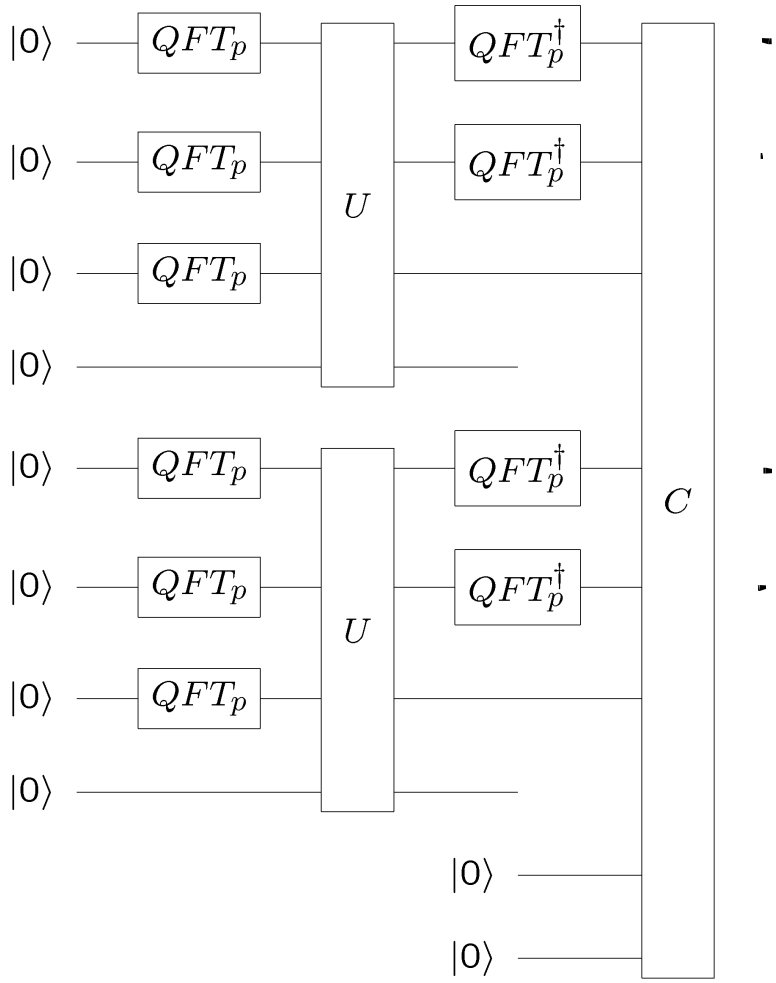
$$\frac{1}{p}\sum_{b_1,b_2\in\mathbb{Z}_p}\omega_p^{vd_1+wd_2}|b_1,b_2\rangle|v,w\rangle \rightarrow \frac{1}{p}|0,0\rangle\sqrt{|S_{v,w}|}\sum_{v,w}\omega_p^{vd_1+wd_2}|v,w\rangle$$

Which has a O(1/2) fidelity squared with the state we desire.

QFTing the second register will let us find d1 and d2

Efficient algorithm for the Heisenberg HSP!

Efficient Quantum Circuit



(Non-Homework) Extensions

This algorithm can be extended to an efficient quantum algorithm for all $\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$

Also, certain metacyclic groups $\mathbb{Z}_N \rtimes \mathbb{Z}_p$

Conclusion

PGM is often g HS states *and* can lead ns.

Efficient algo neralizations.

